

CSA0404 OPERATING SYSTEM

LAB PROGRAM

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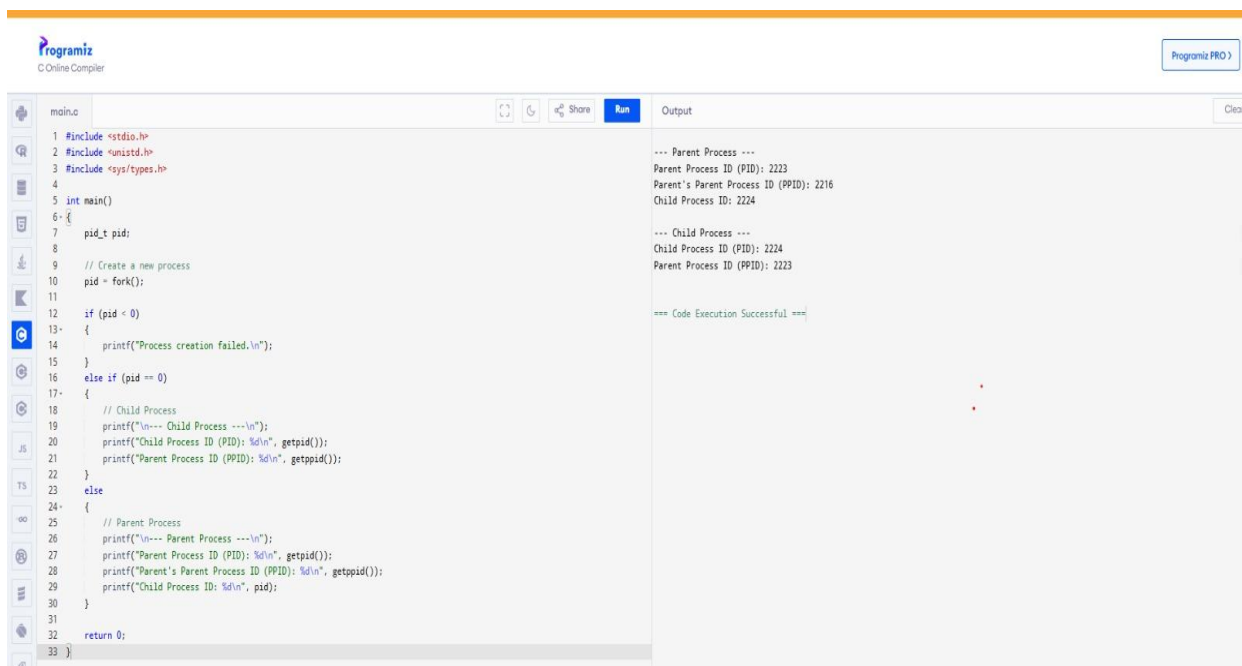
REG NO:192565003

PROGRAMS:

QUESTION 1

Create a new process by invoking the appropriate system call. Get the process identifier of the currently running process and its respective parent using system calls and display the same using a C program.

CODING :



```
1 #include <stdio.h>
2 #include <unistd.h>
3 #include <sys/types.h>
4
5 int main()
6 {
7     pid_t pid;
8
9     // Create a new process
10    pid = fork();
11
12    if (pid < 0)
13    {
14        printf("Process creation failed.\n");
15    }
16    else if (pid == 0)
17    {
18        // Child Process
19        printf("\n--- Child Process ---\n");
20        printf("Child Process ID (PID): %d\n", getpid());
21        printf("Parent Process ID (PPID): %d\n", getppid());
22    }
23    else
24    {
25        // Parent Process
26        printf("\n--- Parent Process ---\n");
27        printf("Parent Process ID (PID): %d\n", getpid());
28        printf("Parent's Parent Process ID (PPID): %d\n", getppid());
29        printf("Child Process ID: %d\n", pid);
30    }
31
32    return 0;
33 }
```

Output

```
--- Parent Process ---
Parent Process ID (PID): 2223
Parent's Parent Process ID (PPID): 2216
Child Process ID: 2224

--- Child Process ---
Child Process ID (PID): 2224
Parent Process ID (PPID): 2223

=== Code Execution Successful ===
```